

# Key distributions

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Some distributions are typically used for outcome models (Table 1); other distributions are typically used for test statistics (Table 2).

Table 1: Distributions typically used for outcome models

| Distribution                      | Uses                                                                                          |
|-----------------------------------|-----------------------------------------------------------------------------------------------|
| <a href="#">Bernoulli</a>         | Binary outcomes                                                                               |
| Binomial                          | Sums of Bernoulli outcomes                                                                    |
| Poisson                           | Unbounded count outcomes                                                                      |
| Geometric                         | Counts of non-events before an event occurs                                                   |
| Negative binomial                 | Mixtures of Poisson distributions, counts of non-events until a given number of events occurs |
| <a href="#">Normal (Gaussian)</a> | Continuous outcomes without a more specific distribution                                      |
| <a href="#">Exponential</a>       | Time to event outcomes                                                                        |
| Gamma                             | Time to event outcomes                                                                        |

| Distribution | Uses                   |
|--------------|------------------------|
| Weibull      | Time to event outcomes |
| Log-normal   | Time to event outcomes |

Table 2: Distributions typically used for test statistics

| Distribution          | Uses                                                                                             |
|-----------------------|--------------------------------------------------------------------------------------------------|
| $\chi^2$              | Regression comparisons (asymptotic), contingency table independence tests, goodness-of-fit tests |
| $F$                   | Gaussian model comparisons (exact)                                                               |
| $Z$ (standard normal) | Proportions, means, regression coefficients (asymptotic)                                         |
| $t$ (Student's)       | Means, regression coefficients in Gaussian outcome models (exact)                                |

## The Bernoulli distribution

**Example 0.1** (A coin flip). The [Bernoulli distribution](#) describes a binary outcome. The indicator of heads on one flip of a fair coin is  $\text{Ber}(1/2)$ . By the [expectation of the Bernoulli distribution](#),  $X \sim \text{Ber}(\pi)$  has mean  $\pi$ , and by the [variance of a Bernoulli random variable](#), it has variance  $\pi(1 - \pi)$ .

## The Poisson distribution

**Definition 0.1** (Poisson distribution). A random variable  $Y$  has the **Poisson distribution** with mean parameter  $\mu > 0$ , written  $Y \sim \text{Pois}(\mu)$ , if:

$$P(Y = y) \stackrel{\text{def}}{=} \frac{\mu^y e^{-\mu}}{y!}, \quad y \in \mathbb{N} = \{0, 1, 2, \dots\} \quad (1)$$

(see Figure 1)

**Exercise 0.1.** What is the range of possible values for a Poisson distribution?

Solution

*Solution 0.1.*

$$\mathcal{R}(Y) = \{0, 1, 2, \dots\} = \mathbb{N}$$

**Theorem 0.1** (CDF of Poisson distribution). *If  $Y \sim \text{Pois}(\mu)$ , then for every real number  $y$ :*

$$P(Y \leq y) = e^{-\mu} \sum_{j=0}^{\lfloor y \rfloor} \frac{\mu^j}{j!} \quad (2)$$

**i** Proof

*Proof.* For  $y < 0$ , both sides are 0 (the sum is empty). For any  $y \geq 0$ , the event  $\{Y \leq y\}$  is the disjoint union of events  $\{Y = j\}$  for all non-negative integers  $j \leq \lfloor y \rfloor$ . Applying the Poisson PMF (Equation 1) and factoring out the common term  $e^{-\mu}$ :

$$\begin{aligned} P(Y \leq y) &= \sum_{j=0}^{\lfloor y \rfloor} P(Y = j) \quad (\text{disjoint union of events } Y = j) \\ &= \sum_{j=0}^{\lfloor y \rfloor} \frac{\mu^j e^{-\mu}}{j!} \quad (\text{definition of Poisson PMF}) \\ &= e^{-\mu} \sum_{j=0}^{\lfloor y \rfloor} \frac{\mu^j}{j!} \quad (\text{factoring out } e^{-\mu} \text{ constant wrt } j) \end{aligned}$$

□

**Example 0.2** (Computing Poisson cumulative probabilities). For a Poisson random variable  $X \sim \text{Pois}(\mu = 2)$ , the probability of observing at most 2 events is computed as:

$$\begin{aligned}
P(X \leq 2) &= e^{-2} \sum_{j=0}^2 \frac{2^j}{j!} && \text{(apply CDF formula with } \mu = 2, y = 2\text{)} \\
&= e^{-2} \left( \frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} \right) && \text{(expand terms for } j = 0, 1, 2\text{)} \\
&= e^{-2}(1 + 2 + 2) && \text{(simplify factorials and powers)} \\
&= 5e^{-2} \approx 0.677 && \text{(evaluate numerical value)}
\end{aligned}$$

(see Figure 2)

```

pois_dists <-
  dplyr::tibble(mu = c(0.5, 1, 2, 5, 10, 20)) |>
  dplyr::reframe(.by = mu, x = 0:30) |>
  dplyr::mutate(
    `P(X = x)` = dpois(x, lambda = mu),
    `P(X <= x)` = ppois(x, lambda = mu),
    mu = factor(mu)
  )

plot0 <-
  pois_dists |>
  ggplot2::ggplot(
    ggplot2::aes(
      x = x,
      y = `P(X = x)`,
      fill = mu,
      col = mu
    )
  ) +
  ggplot2::theme(legend.position = "bottom") +
  ggplot2::labs(
    fill = latex2exp::TeX("$\\mu$"),
    col = latex2exp::TeX("$\\mu$"),
    y = latex2exp::TeX("$\\Pr_{\\mu}(X = x)$")
  )

plot1 <-
  plot0 +
  ggplot2::geom_segment(yend = 0) +
  ggplot2::facet_wrap(~mu)

print(plot1)

```

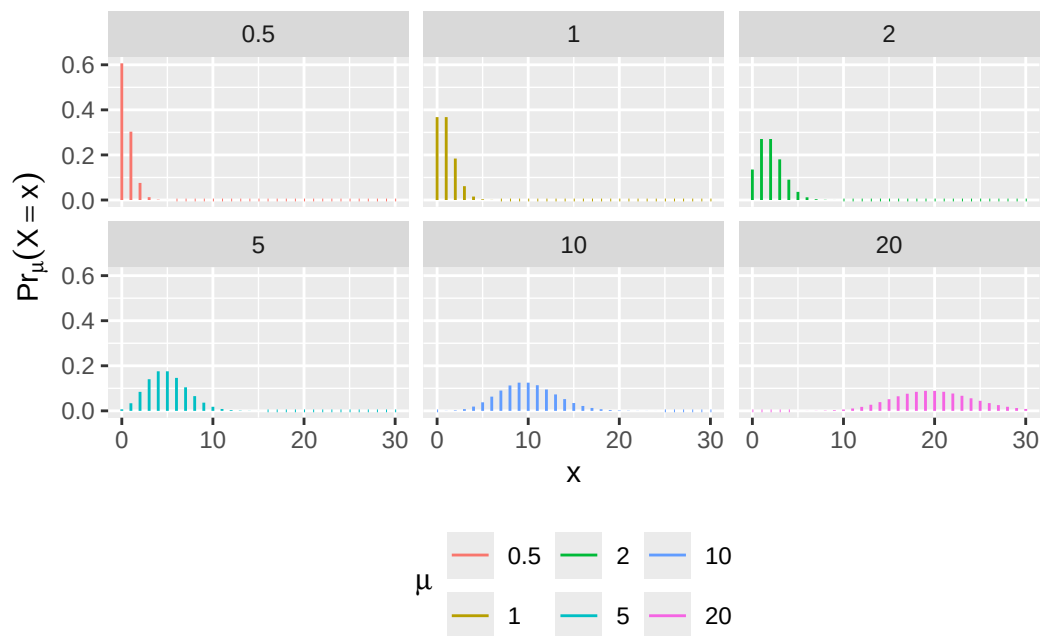


Figure 1: Poisson PMFs, by mean parameter  $\mu$

```
plot2 <-
  plot0 +
  ggplot2::geom_step(alpha = 0.75) +
  ggplot2::aes(y = `P(X <= x)`) +
  ggplot2::labs(y = latex2exp::TeX("$\\Pr_{\\mu}(X \\leq x)$"))

print(plot2)
```



Figure 2: Poisson CDFs

**Exercise 0.2** (Poisson distribution functions). Let  $X \sim \text{Pois}(\mu = 3.75)$ . Compute:

- $P(X = 4)$
- $P(X \leq 7)$
- $P(X > 5)$

Solution

*Solution.*

- $P(X = 4) = 0.1937803$
- $P(X \leq 7) = 0.9623787$
- $P(X > 5) = 0.1771172$

**Theorem 0.2** (Properties of the Poisson distribution). If  $X \sim \text{Pois}(\mu)$ , then:

- $E[X] = \mu$
- $\text{Var}(X) = \mu$
- $P(X = x) = \frac{\mu}{x} P(X = x - 1)$  for  $x \in \{1, 2, \dots\}$

- For  $x \in \{1, 2, \dots\}$  with  $x < \mu$ ,  $P(X = x) > P(X = x - 1)$
- For  $x = \mu$  (possible only when  $\mu$  is an integer),  $P(X = x) = P(X = x - 1)$
- For  $x \in \{1, 2, \dots\}$  with  $x > \mu$ ,  $P(X = x) < P(X = x - 1)$
- If  $\mu$  is not an integer,  $\arg \max_x P(X = x) = \lfloor \mu \rfloor$ ; if  $\mu$  is an integer, the maximum is attained at both  $x = \mu - 1$  and  $x = \mu$

## i Proof

*Proof.* **Mean.**

$$\begin{aligned}
 E[X] &= \sum_{x=0}^{\infty} x \cdot P(X = x) && \text{(definition of expected value)} \\
 &= 0 \cdot P(X = 0) + \sum_{x=1}^{\infty} x \cdot P(X = x) && \text{(separate } x = 0 \text{ term)} \\
 &= \sum_{x=1}^{\infty} x \cdot \frac{\mu^x e^{-\mu}}{x!} && \text{(substitute Poisson PMF)} \\
 &= \sum_{x=1}^{\infty} x \cdot \frac{\mu^x e^{-\mu}}{x \cdot (x-1)!} && \text{(definition of factorial } x!) \\
 &= \sum_{x=1}^{\infty} \frac{\mu^x e^{-\mu}}{(x-1)!} && \text{(cancel factor of } x) \\
 &= \mu \cdot \sum_{x=1}^{\infty} \frac{\mu^{x-1} e^{-\mu}}{(x-1)!} && \text{(factor out one power of } \mu) \\
 &= \mu \cdot \sum_{y=0}^{\infty} \frac{\mu^y e^{-\mu}}{y!} && \text{(change index variable } y \stackrel{\text{def}}{=} x - 1) \\
 &= \mu \cdot 1 && \text{(PMF sums to 1 over state space)} \\
 &= \mu && \text{(simplify)}
 \end{aligned}$$

**Variance.** The same steps, canceling two factors instead of one, give  $E[X(X-1)]$ :

$$\begin{aligned}
 E[X(X-1)] &= \sum_{x=2}^{\infty} x(x-1) \cdot \frac{\mu^x e^{-\mu}}{x!} && \text{(LOTUS; the } x = 0, 1 \text{ terms are 0)} \\
 &= \sum_{x=2}^{\infty} \frac{\mu^x e^{-\mu}}{(x-2)!} && \text{(cancel } x(x-1) \text{ against } x!) \\
 &= \mu^2 \cdot \sum_{y=0}^{\infty} \frac{\mu^y e^{-\mu}}{y!} && \text{(factor out } \mu^2; y \stackrel{\text{def}}{=} x - 2) \\
 &= \mu^2 && \text{(PMF sums to 1)}
 \end{aligned}$$

Then, by the [simplified expression for variance](#) and [linearity of expectation](#):

$$\begin{aligned}
\text{Var}(X) &= E[X^2] - (E[X])^2 && \text{(simplified expression for variance)} \\
&= E[X(X-1)] + E[X] - (E[X])^2 && (X^2 = X(X-1) + X; \text{ linearity}) \\
&= \mu^2 + \mu - \mu^2 && \text{(substitute)} \\
&= \mu && \text{(simplify)}
\end{aligned}$$

**Ratio of consecutive probabilities.** For  $x \in \{1, 2, \dots\}$ :

$$\begin{aligned}
\frac{P(X = x)}{P(X = x-1)} &= \frac{\mu^x e^{-\mu} / x!}{\mu^{x-1} e^{-\mu} / (x-1)!} && \text{(substitute Poisson PMF)} \\
&= \frac{\mu}{x} && \text{(cancel } \mu^{x-1} e^{-\mu} \text{ and } (x-1)!)
\end{aligned}$$

This ratio is greater than, equal to, or less than 1 as  $x$  is less than, equal to, or greater than  $\mu$ , which gives the three comparisons. So  $P(X = x)$  increases while  $x < \mu$  and decreases once  $x > \mu$ . If  $\mu$  is not an integer, the last increase is at  $x = \lfloor \mu \rfloor$ , which is therefore the unique mode. If  $\mu$  is an integer,  $P(X = \mu) = P(X = \mu - 1)$ , and both are modes.

See also <https://statproofbook.github.io/P/poiss-mean> and <https://statproofbook.github.io/P/poiss-var>.  $\square$

**Example 0.3** (Mode of a Poisson distribution). For  $X \sim \text{Pois}(2.5)$ ,  $P(X = 2) = \frac{2.5}{2} P(X = 1) > P(X = 1)$  and  $P(X = 3) = \frac{2.5}{3} P(X = 2) < P(X = 2)$ , so the mode is  $\lfloor 2.5 \rfloor = 2$ . For  $X \sim \text{Pois}(2)$ ,  $P(X = 2) = \frac{2}{2} P(X = 1)$ , so 1 and 2 are both modes, each with probability  $2e^{-2} \approx 0.271$ .

**Definition 0.2** (Exposure magnitude). For many count outcomes, there is some sense of an **exposure magnitude**, such as **population size** or **duration of observation**, which multiplicatively rescales the expected (mean) count.

**Exercise 0.3.** What are some examples of exposure magnitudes?

Solution

*Solution.*

Table 3: Examples of exposure units

| outcome           | exposure units                              |
|-------------------|---------------------------------------------|
| disease incidence | number of individuals exposed; time at risk |

car accidents  
 worksite accidents  
 population size

miles driven  
 person-hours worked  
 size of habitat

Exposure units are similar to the number of trials in a binomial distribution, but **in non-binomial count outcomes, there can be more than one event per unit of exposure.**

We can use  $t$  to represent continuous-valued exposures/observation durations, and  $n$  to represent discrete-valued exposures.

**Definition 0.3** (Event rate). For a count  $Y$  observed over a fixed, known **exposure magnitude**  $t > 0$ , the **event rate**, denoted  $\lambda$ , is the mean of  $Y$  divided by the exposure magnitude:

$$\lambda \stackrel{\text{def}}{=} \frac{E[Y]}{t} \quad (3)$$

The event rate is a transformation of the mean: it removes the exposure magnitude from the mean, so counts observed over different exposures can be compared on one scale. Regression models for counts use the same decomposition, with the rate depending on covariates and the exposure magnitude known (see [rme's count-regression chapter](#)).

**Theorem 0.3** (Transformation function from event rate to mean). *If a count  $Y$  is observed over exposure magnitude  $t > 0$  with event rate  $\lambda$ , then its mean  $\mu \stackrel{\text{def}}{=} E[Y]$  is:*

$$\mu = \lambda \cdot t \quad (4)$$

#### **i** Proof

*Proof.* By Definition 0.3:

$$\begin{aligned} \lambda &\stackrel{\text{def}}{=} \frac{E[Y]}{t} && \text{(definition of event rate)} \\ E[Y] &= \lambda \cdot t && \text{(multiply both sides by } t > 0) \\ \mu &= \lambda \cdot t && (\mu \stackrel{\text{def}}{=} E[Y]) \end{aligned}$$

□

**Example 0.4** (Calculating expected counts from event rates). Suppose a city records a disease event rate of  $\lambda = 0.05$  cases per person-year. For a subpopulation with an exposure magnitude of  $t = 100$  person-years, the expected count of cases is, by Theorem 0.3:

$$\begin{aligned}
\mu &= \lambda \cdot t && \text{(transformation from event rate to mean)} \\
&= 0.05 \times 100 && \text{(substitute } \lambda = 0.05 \text{ and } t = 100) \\
&= 5 \text{ cases} && \text{(evaluate expected count)}
\end{aligned}$$

**Theorem 0.4** (No exposure means no expected events). *For each exposure magnitude  $t \geq 0$ , let  $Y_t$  be the count observed over exposure  $t$ . If the mean count is proportional to the exposure magnitude,  $E[Y_t] = \lambda \cdot t$  for all  $t \geq 0$  with one finite rate  $\lambda$ , then there are no expected events without exposure:*

$$E[Y_0] = 0$$

#### **i** Proof

*Proof.*

$$\begin{aligned}
E[Y_0] &= \lambda \cdot 0 && \text{(evaluate } E[Y_t] = \lambda t \text{ at } t = 0) \\
&= 0 && \text{(multiplication by zero; } \lambda \text{ is finite)}
\end{aligned}$$

□

The hypothesis carries the content here. Definition 0.3 alone cannot give  $E[Y_0] = 0$ , since  $\lambda = E[Y]/t$  is undefined at  $t = 0$ ; the result holds for a model that assumes one rate  $\lambda$  shared across exposure magnitudes, including  $t = 0$ .

**Example 0.5** (Zero exposure time). If a subject is observed for  $t = 0$  person-years, no follow-up time has elapsed, so under a constant-rate model the expected number of incident events is  $E[Y_0] = 0$ .

#### **!** Important

A mean proportional to exposure,  $E[Y_t] = \lambda t$ , has no term that stays nonzero at  $t = 0$ : a model of this form says that with no exposure, no events are expected. A mean with an added constant, such as  $E[Y_t] = c + \lambda t$  with  $c > 0$ , would expect  $c$  events even with no exposure. Regression models for counts keep the proportional form when they add covariates (see [rme's count-regression chapter](#)).

**Theorem 0.5** (Exposure is additive on the log scale). *If  $\mu = \lambda \cdot t$  with  $\lambda > 0$  and  $t > 0$ , then:*

$$\log \mu = \log \lambda + \log t$$

**i** Proof

*Proof.*

$$\begin{aligned}\log \mu &= \log(\lambda \cdot t) && \text{(substitute } \mu = \lambda \cdot t \text{)} \\ &= \log \lambda + \log t && \text{(logarithm product rule)}\end{aligned}$$

□

**Example 0.6** (Log-linear representation of expected counts). If a clinic sees an event rate of  $\lambda = 0.02$  events/day and  $t = 30$  days of observation:

$$\begin{aligned}\log \mu &= \log(0.02) + \log(30) && \text{(apply log-scale formula)} \\ &= -3.912 + 3.401 && \text{(evaluate natural logarithms)} \\ &= -0.511 && \text{(sum terms)}\end{aligned}$$

Exponentiating yields  $\mu = \exp\{-0.511\} \approx 0.60$  expected events.

**Definition 0.4** (Offset). For a count outcome with exposure magnitude  $t$ , the known term  $\log t$  in the log-scale decomposition of the mean (Theorem 0.5),

$$\log \mu = \log \lambda + \log t,$$

is called the **offset**: it shifts  $\log \mu$  by a known amount, with no unknown coefficient to estimate.

The offset needs no covariates: with a single exposure  $t$  and an unknown rate  $\lambda$ ,  $\log t$  is already an offset. Regression models for counts keep the same term and add covariate terms beside it (see [rme's count-regression chapter](#)).

**Example 0.7** (The offset for a clinic's event count). In Example 0.6, the clinic was observed for  $t = 30$  days, so the offset is  $\log 30 \approx 3.401$ . Only  $\log \lambda$  is unknown before the data are seen; the offset is fixed by the length of observation. A clinic observed for  $t = 60$  days has offset  $\log 60 \approx 4.094$ , which raises  $\log \mu$  by  $\log 2 \approx 0.693$  at the same event rate.

**Theorem 0.6** (Sum of independent Poisson random variables). *If  $X$  and  $Y$  are [independent](#) Poisson random variables with means  $\mu_X$  and  $\mu_Y$ , their sum,  $Z = X + Y$ , is also a Poisson random variable, with mean  $\mu_Z = \mu_X + \mu_Y$ .*

### i Proof

*Proof.* The event  $\{Z = z\}$  is the disjoint union of the events  $\{X = k, Y = z - k\}$  for  $k = 0, 1, \dots, z$  (since  $X$  and  $Y$  are non-negative integers), so:

$$\begin{aligned}
 P(Z = z) &= \sum_{k=0}^z P(X = k, Y = z - k) && \text{(countable additivity)} \\
 &= \sum_{k=0}^z P(X = k) P(Y = z - k) && \text{(independence)} \\
 &= \sum_{k=0}^z \frac{\mu_X^k e^{-\mu_X}}{k!} \frac{\mu_Y^{z-k} e^{-\mu_Y}}{(z-k)!} && \text{(substitute Poisson PMFs)} \\
 &= e^{-(\mu_X + \mu_Y)} \sum_{k=0}^z \frac{\mu_X^k \mu_Y^{z-k}}{k! (z-k)!} && \text{(factor out } e^{-\mu_X} e^{-\mu_Y} \text{)} \\
 &= \frac{e^{-(\mu_X + \mu_Y)}}{z!} \sum_{k=0}^z \frac{z!}{k! (z-k)!} \mu_X^k \mu_Y^{z-k} && \text{(multiply and divide by } z! \text{)} \\
 &= \frac{e^{-(\mu_X + \mu_Y)}}{z!} (\mu_X + \mu_Y)^z && \text{(binomial theorem)}
 \end{aligned}$$

This probability expression matches the PMF of a  $\text{Pois}(\mu_X + \mu_Y)$  random variable (see also [https://web.stanford.edu/class/archive/cs/cs109/cs109.1206/lectureNotes/LN12\\_independent\\_rvs.pdf](https://web.stanford.edu/class/archive/cs/cs109/cs109.1206/lectureNotes/LN12_independent_rvs.pdf), Example 3).  $\square$

**Example 0.8** (Aggregating independent region counts). Suppose Region A records  $X \sim \text{Pois}(\mu_X = 12)$  cases and Region B records  $Y \sim \text{Pois}(\mu_Y = 18)$  cases independently. By Theorem 0.6, the combined total count  $Z = X + Y$  follows a Poisson distribution:

$$\begin{aligned}
 Z &\sim \text{Pois}(\mu_X + \mu_Y) && \text{(sum of independent Poissons)} \\
 &= \text{Pois}(12 + 18) && \text{(substitute region means)} \\
 &= \text{Pois}(30) && \text{(evaluate sum)}
 \end{aligned}$$

## The Negative-Binomial distribution

**Definition 0.5** (Negative binomial distribution). A random variable  $Y$  has the **negative binomial distribution** with mean  $\mu > 0$  and overdispersion parameter  $\rho > 0$ , written  $Y \sim \text{NegBin}(\mu, \rho)$ , if, for  $y \in \{0, 1, 2, \dots\}$ :

$$P(Y = y) \stackrel{\text{def}}{=} \frac{\mu^y}{y!} \cdot \frac{\Gamma(\rho + y)}{\Gamma(\rho) \cdot (\rho + \mu)^y} \cdot \left(1 + \frac{\mu}{\rho}\right)^{-\rho}$$

where  $\Gamma$  is the gamma function, which satisfies  $\Gamma(x) = (x-1)!$  for positive integers  $x$ .

As  $\rho \rightarrow \infty$ , the second factor converges to 1 and the third factor converges to  $\exp\{-\mu\}$ , which brings us back to the Poisson distribution.

**Theorem 0.7** (Mean and variance of the negative binomial distribution). *If  $Y \sim \text{NegBin}(\mu, \rho)$ , then:*

- $E[Y] = \mu$
- $\text{Var}(Y) = \mu + \frac{\mu^2}{\rho} > \mu$

### **i** Proof

*Proof.* The negative binomial distribution is a gamma mixture of Poisson distributions: if  $\Lambda$  has the gamma density  $g(\lambda) = \frac{(\rho/\mu)^\rho}{\Gamma(\rho)} \lambda^{\rho-1} e^{-\rho\lambda/\mu}$  for  $\lambda > 0$ , which has mean  $\mu$  and variance  $\mu^2/\rho$  (Casella and Berger 2002), and  $Y \mid \Lambda = \lambda \sim \text{Pois}(\lambda)$ , then  $Y \sim \text{NegBin}(\mu, \rho)$ . To check this, integrate the joint density over  $\lambda$ :

$$\begin{aligned}
 P(Y = y) &= \int_0^\infty \frac{\lambda^y e^{-\lambda}}{y!} \cdot \frac{(\rho/\mu)^\rho}{\Gamma(\rho)} \lambda^{\rho-1} e^{-\rho\lambda/\mu} d\lambda \quad (\text{marginalize over } \Lambda) \\
 &= \frac{(\rho/\mu)^\rho}{y! \Gamma(\rho)} \int_0^\infty \lambda^{y+\rho-1} e^{-\lambda(1+\rho/\mu)} d\lambda \quad (\text{collect powers of } \lambda \text{ and exponents}) \\
 &= \frac{(\rho/\mu)^\rho}{y! \Gamma(\rho)} \cdot \frac{\Gamma(y+\rho)}{(1+\rho/\mu)^{y+\rho}} \quad (\int_0^\infty \lambda^{a-1} e^{-b\lambda} d\lambda = \Gamma(a)/b^a) \\
 &= \frac{\Gamma(y+\rho)}{y! \Gamma(\rho)} \left(\frac{\rho}{\mu}\right)^\rho \left(\frac{\mu}{\mu+\rho}\right)^{y+\rho} \quad (1+\rho/\mu = (\mu+\rho)/\mu) \\
 &= \frac{\Gamma(y+\rho)}{y! \Gamma(\rho)} \left(\frac{\rho}{\mu+\rho}\right)^\rho \left(\frac{\mu}{\mu+\rho}\right)^y \quad (\text{combine the } \mu^\rho \text{ factors}) \\
 &= \frac{\mu^y}{y!} \cdot \frac{\Gamma(\rho+y)}{\Gamma(\rho) (\rho+\mu)^y} \cdot \left(1 + \frac{\mu}{\rho}\right)^{-\rho} \quad (\text{rearrange into the form of the definition})
 \end{aligned}$$

Then, by the [law of iterated expectations](#) and the [law of total variance](#), using  $E[Y \mid \Lambda] = \text{Var}(Y \mid \Lambda) = \Lambda$  (Theorem 0.2):

$$\begin{aligned}
E[Y] &= E[E[Y \mid \Lambda]] && \text{(law of iterated expectations)} \\
&= E[\Lambda] && (E[Y \mid \Lambda] = \Lambda) \\
&= \mu && \text{(mean of the gamma distribution)} \\
\text{Var}(Y) &= E[\text{Var}(Y \mid \Lambda)] + \text{Var}(E[Y \mid \Lambda]) && \text{(law of total variance)} \\
&= E[\Lambda] + \text{Var}(\Lambda) && (\text{Var}(Y \mid \Lambda) = E[Y \mid \Lambda] = \Lambda) \\
&= \mu + \frac{\mu^2}{\rho} && \text{(mean and variance of the gamma distribution)}
\end{aligned}$$

and  $\mu^2/\rho > 0$  gives  $\text{Var}(Y) > \mu$ . □

**Example 0.9** (Overdispersion relative to the Poisson). With  $\mu = 4$  and  $\rho = 2$ ,  $\text{Var}(Y) = 4 + 16/2 = 12$ , three times the variance of a  $\text{Pois}(4)$  count with the same mean.

## Weibull distribution

**Definition 0.6** (Weibull distribution). A non-negative random variable  $T$  has the **Weibull distribution** with shape  $\alpha > 0$  and rate  $\lambda > 0$  if its [survival function](#) is:

$$S(t) \stackrel{\text{def}}{=} e^{-\lambda t^\alpha}, \quad t \geq 0$$

**Theorem 0.8** (Weibull density, hazard, and mean). *If  $T$  has the Weibull distribution with shape  $\alpha$  and rate  $\lambda$ , then for  $t > 0$ :*

$$\begin{aligned}
f(t) &= \alpha \lambda t^{\alpha-1} e^{-\lambda t^\alpha} \\
h(t) &= \alpha \lambda t^{\alpha-1} \\
E[T] &= \Gamma(1 + 1/\alpha) \cdot \lambda^{-1/\alpha}
\end{aligned}$$

### Proof

*Proof.* The CDF of  $T$  is  $F(t) = 1 - S(t)$  ([survival function and CDF](#)), which is 0 for  $t < 0$  and  $1 - e^{-\lambda t^\alpha}$  for  $t \geq 0$ . This  $F$  is continuous everywhere, with a continuous derivative everywhere except possibly  $t = 0$ , so  $f = F' = -S'$  is a density of  $T$  ([a piecewise-smooth CDF has its derivative as a density](#)). This density is continuous at every  $t > 0$ , so there the hazard is  $f(t)/S(t)$  ([hazard equals density over survival](#)):

$$\begin{aligned}
f(t) &= -\frac{d}{dt}e^{-\lambda t^\alpha} && (f = -S') \\
&= \alpha \lambda t^{\alpha-1} e^{-\lambda t^\alpha} && (\text{chain rule}) \\
h(t) &= \frac{\alpha \lambda t^{\alpha-1} e^{-\lambda t^\alpha}}{e^{-\lambda t^\alpha}} && (\text{hazard is density over survival}) \\
&= \alpha \lambda t^{\alpha-1} && (\text{cancel})
\end{aligned}$$

For the mean, use the [survival-function formula for the mean](#) and substitute  $u = \lambda t^\alpha$ , so that  $t = (u/\lambda)^{1/\alpha}$  and  $dt = \frac{1}{\alpha} \lambda^{-1/\alpha} u^{1/\alpha-1} du$ :

$$\begin{aligned}
E[T] &= \int_0^\infty e^{-\lambda t^\alpha} dt && (\text{expectation via the survival function}) \\
&= \frac{1}{\alpha} \lambda^{-1/\alpha} \int_0^\infty u^{1/\alpha-1} e^{-u} du && (\text{substitute } u = \lambda t^\alpha) \\
&= \frac{1}{\alpha} \lambda^{-1/\alpha} \Gamma(1/\alpha) && (\text{definition of the gamma function}) \\
&= \Gamma(1 + 1/\alpha) \lambda^{-1/\alpha} && (\Gamma(1 + a) = a \Gamma(a))
\end{aligned}$$

□

The hazard is written  $h(t)$  here, rather than  $\lambda(t)$ , because the Weibull rate parameter is also called  $\lambda$ .

When  $\alpha = 1$ , the Weibull distribution reduces to the exponential distribution. When  $\alpha > 1$  the hazard is increasing and when  $\alpha < 1$  the hazard is decreasing. This property provides more flexibility than the exponential.

**Example 0.10** (Exponential as a special case). With  $\alpha = 1$ , Theorem [0.8](#) gives  $h(t) = \lambda$  and  $E[T] = \Gamma(2)\lambda^{-1} = 1/\lambda$ , matching the exponential distribution's constant hazard and mean. With  $\alpha = 2$  and  $\lambda = 1$ ,  $h(t) = 2t$  increases with  $t$ , and  $E[T] = \Gamma(3/2) = \sqrt{\pi}/2 \approx 0.886$ .

## References

Casella, George, and Roger Berger. 2002. *Statistical Inference*. 2nd ed. Cengage Learning. <https://www.cengage.com/c/statistical-inference-2e-casella-berger/9780534243128/>.